ChatGPT Auto

how derivative and integration cancels out eachother like derivative to integration or integration to derivative

Derivatives and integrals **"cancel each other"** because they are **inverse operations**—similar to how **multiplication and division** undo each other. This relationship is explained by the **Fundamental Theorem of Calculus**. Let's see both directions.

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1. Derivative of an Integral

If you integrate a function and then take its derivative, you get the original function back.

$$\frac{d}{dx}\left(\int f(x)\,dx\right) = f(x)$$

Example

$$\int x^2\,dx = \frac{x^3}{3} + C$$

Now differentiate:

$$\frac{d}{dx}\left(\frac{x^3}{3} + C\right) = x^2$$

So:

$$\frac{d}{dx}\left(\int x^2 dx\right) = x^2$$

✔ The derivative **cancels the integral**.

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2. Integral of a Derivative

If you differentiate a function and then integrate it, you get the original function **plus a constant**.

$$\int \frac{d}{dx} f(x)\,dx = f(x) + C$$

Example

Start with

$$f(x) = x^3$$

Derivative:

$$\frac{d}{dx}(x^3) = 3x^2$$

Now integrate:

$$\int 3x^2 dx = x^3 + C$$

✔ We get the original function **with a constant**.

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3. Why the Constant Appears

When differentiating:

$$\frac{d}{dx}(x^3 + 5) = 3x^2$$

$$\frac{d}{dx}(x^3 + 100) = 3x^2$$

Different functions give the **same derivative**.

So when integrating, we must add **+C** because we cannot know the lost constant.

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4. Intuition (Simple Idea)

Think of:

- **Derivative** → slope / rate of change 📈
- **Integral** → accumulated area 📊

Derivative **breaks a function into tiny changes**, while integration **adds those tiny changes back together**.

So they undo each other.

✔ **Summary**

Operation	Result
$\frac{d}{dx}(\int f(x)dx)$	$f(x)$
$\int \frac{d}{dx} f(x)dx$	$f(x)+C$

If you want, I can also show the **visual intuition (area vs slope)** that makes this much easier to understand. It's the way professors usually explain it. 📉📈

area under the curve means how find also let me know also let me know if we convert derivative into integral and integral into derivative

1. What does "Area under the curve" mean?

When you draw a graph of a function $y = f(x)$, the **area under the curve** means the region between:

- the **curve**
- the **x-axis**
- two x-values (for example a and b)

Mathematically:

$$\text{Area} = \int_a^b f(x)\,dx$$

This is called a **definite integral**.

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2. How to find area under the curve (example)

Suppose

$$f(x) = x^2$$

Find area from **0 to 2**.

Step 1: Integrate the function

$$\int x^2 dx = \frac{x^3}{3}$$

Step 2: Apply limits

$$\left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{2^3}{3} - \frac{0^3}{3}$$

$$= \frac{8}{3}$$

✅ Area under the curve = **8/3 square units**

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3. Converting Derivative → Integral

If you have a derivative:

$$f'(x) = 6x$$

Convert to function using integration:

$$f(x) = \int 6x \, dx$$

$$f(x) = 3x^2 + C$$

So integration **undoes differentiation**.

4. Converting Integral → Derivative

Example:

$$F(x) = \int x^3 dx$$

First integrate:

$$F(x) = \frac{x^4}{4} + C$$

Now differentiate:

$$\frac{d}{dx}\left(\frac{x^4}{4} + C\right) = x^3$$

So derivative **undoes integration**.

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5. Simple idea to remember

Concept	Meaning
Derivative	Slope / rate of change
Integral	Area accumulated under curve

They are **inverse operations** (Fundamental Theorem of Calculus).

✅ Key formulas

Derivative of integral:

$$\frac{d}{dx}\left(\int f(x)dx\right) = f(x)$$

Integral of derivative:

$$\int f'(x)dx = f(x) + C$$

If you want, I can also explain **why integral gives area under the curve (with a very simple rectangle method that makes everything click).**

